CMP3060 — Computer Graphics

OCEAN ADVENTURE
Grand Tide

A 3D Survival Game — Defend Your Raft, Survive the Sea

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Game Proposal | Deadline: Monday, 27 April 2026

Game Overview

Game Description

Ocean Adventure: Grand Tide is a fast-paced 3D survival game in which the player takes on the role of a castaway stranded on a drifting wooden raft in the middle of a vast, dangerous ocean.

The primary objective is to **survive as long as possible** by defending the raft against waves of dynamically spawning enemy marine ships and floating ocean hazards. The player wields a **magical fire-based combat system** — aiming and launching projectile attacks to destroy enemy vessels before they can ram into and destroy the raft.

As the player's survival score increases, enemy waves grow faster and more numerous, escalating the tension with every passing moment. The world is brought to life through a **full day-and-night cycle** illuminated by dynamic lights, fire-glow from projectiles, and explosion flares that cast real-time colour onto the surrounding water and raft geometry.

Genre:	3D Survival / Arcade	**Camera:**	Third-Person Dynamic
Win Goal:	Max survival time	**Core Loop:**	Aim → Shoot → Survive

Concept Art / Mockup

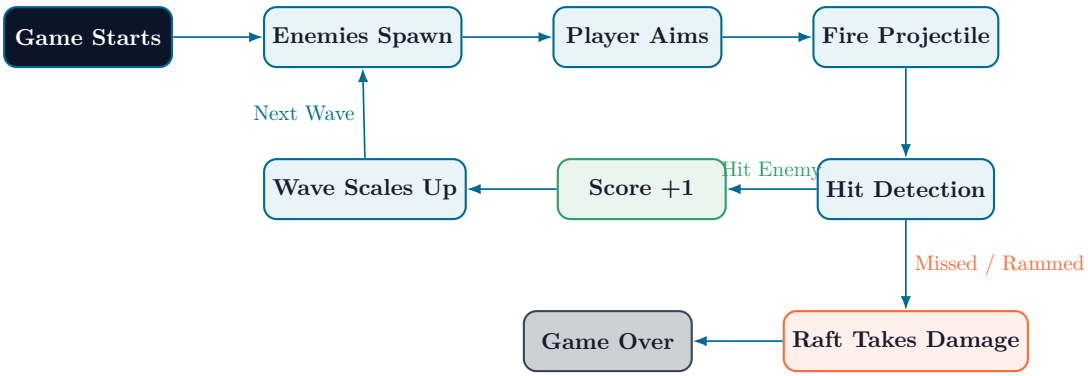
Figure 1 — Ocean Adventure: Grand Tide — Game Mockup

Technical Alignment with Requirements

The table below maps each required feature to our planned implementation.

#	Requirement	Implementation Plan
	3D Models & Textures	Imported meshes for player character, wooden raft, enemy ships, floating barrels and ocean surface. All objects use **lit materials** with albedo, specular, and roughness texture maps.
	Lighting	**Directional light** for sun / moon (day-night cycle). **Dynamic point lights** emitted by fire projectiles and ship explosions that illuminate nearby geometry and water in real time.
	Dynamic Elements	Enemy ships **spawn dynamically** and navigate toward the player's raft. Floating barrel hazards drift across the scene. Every object reacts to player actions.
	Interaction & Game Logic	Core loop: *aim* with the cursor, *shoot* fire projectiles, *score points* per destroyed ship. Difficulty scales with the survival timer. Clear win/lose conditions are defined.
	Collision Detection	**3D bounding-box collision** for ship-raft impacts; **ray-sphere casting** for projectile-to-ship hit registration.
	Camera Motion	**Third-person dynamic camera** follows the player, pans with aiming input, and smoothly adjusts height/distance during combat.
	Sky Sphere	Full **sky sphere** with a seamless ocean-sky panoramic texture. Blends between day and night sky states as the in-game time progresses.
	Post-Processing	Custom **screen-shake** + **red-vignette pulse** effect triggered on raft damage. Optionally, a **chromatic aberration** burst on explosions — both implemented via framebuffer post-processing.
	Game States	**Menu State** (title screen, start/quit buttons) and **Play State** (active gameplay), with clean transitions. All scene data loaded from `.jsonc` config files — **no hard-coded scene data**.

Core Gameplay Loop



Prototype-First Development Strategy

1. **Minimal Prototype:** Single raft mesh, one enemy ship that moves toward the player, basic fire projectile with ray-sphere hit detection. Verify all Phase 1 rendering systems work together.
2. **Core Mechanics:** Add spawn manager (waves), score counter, raft HP, and game-over condition.
3. **Polish:** Integrate lit materials, day-night directional light, dynamic point lights from fire/explosions, sky sphere, and post-processing effects.
4. **Content Expansion:** Full ocean environment, all 3D models, complete config-file scene descriptions, and menu state integration.